

7 (short)

(a) Explain what is meant by static and dynamic hazards. [2]

(b) Use a Karnaugh map to simplify the Boolean expression below.

$$A\bar{B}.C.D.\bar{E} + A\bar{B}.\bar{C}.D.\bar{E} + A\bar{B}.\bar{C}.D.E + A.B.\bar{C}.D.E$$

Draw a circuit that implements this by means of NAND gates and inverters only, using the least number of these components possible. [6]

(c) Explain how you would modify your simplified expression in (b) such that static 1-hazards are also avoided. [2]